GNN-based SHM for H3 Rocket Payload Fairing

5-class defect detection + PINN regularisation + multitask centroid regression

Keisuke Nishioka

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Institut für Kontinuumsmechanik, LUH

Outline

Problem setup

SHM problem: two complementary analyses

Structure: CFRP / Al-honeycomb sandwich fairing (H3 Type-S, 1/6 section) **Defects:** debonding / FOD / impact / delamination

(A) Static FEM analysis

- Abaqus static stress field per defect scenario
- **10,897 nodes × 34 features** (disp, stress, temp, layup, . . .)
- Graph: FEM connectivity
- Task: **5-class node classification** + defect centroid regression
- 700 simulations (560/140 split)
- Metric: OptF1 (threshold-optimal)

(B) Dynamic guided-wave (GW) analysis

- Abaqus explicit GW propagation (piezo excitation)
- **9 sensors × time series** (sampled at 4 MHz)
- Graph: sensor spatial network
- Task: **binary defect detection** + wave-phase PINN (Level 3)
- Dataset: 108 samples (101 healthy / 7 defect, 9-sensor consistent)
- Model: MIFNO-GW (2-D factorised FNO, waveform-only)

Node features (34 dims)

Dims	Feature group	Physical meaning
0–2	Position (x, y, z)	Node coordinates
3–5	Normal (n_x, n_y, n_z)	Surface orientation
6–9	Curvatures (k_1, k_2, H, K)	Shell geometry
10–13	Displacement $(u_x, u_y, u_z, \|u\|)$	Structural response
14	Temperature	Thermal loading
15–18	Stress $(s_{11}, s_{22}, s_{12}, \sigma_{\text{Mises}})$	Failure indicators
19–23	Principal stress, thermal σ_M, log-strain	Derived quantities
24–30	Fibre direction, layup (0°/45°/−45°/90°)	Anisotropy
31–33	Circumferential angle, boundary/load flags	Topology

Methods

Three modelling additions

(A) Multi-task learning — defect centroid regression

Joint loss: $\mathcal{L} = \mathcal{L}_{\text{focal}} + \lambda_{\text{reg}}\, \mathcal{L}_{\text{reg}}$

Shared backbone $\longrightarrow$ node classification head + graph-level MLP for (x, y, z) centroid.

GT centroid computed on-the-fly from defective nodes.

(B) Level-1 PINN — stress gradient regularisation

High predicted defect probability should co-locate with high $|\nabla \sigma_{\text{Mises}}|$.

$\mathcal{L}_{\text{stress}} = \frac{1}{N} \sum_i p_i^{(\text{def})} \exp(-|\Delta\sigma_i|/\sigma_0)$

(C) Level-2 PINN — static equilibrium residual $\nabla\cdot\sigma = 0$

Graph-edge finite differences on (s_{11}, s_{22}, s_{12}); penalises defect predictions at nodes with low equilibrium violation.

LGSTA — Local-Global Attention GNN

Best-performing architecture: LocalGlobalAttentionGNN (inspired by LGSTA-GNN, Buildings 2025)

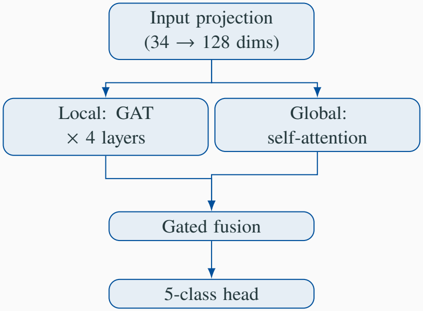
Local branch (GAT $\times$ 4 layers)

- Neighbourhood message-passing
- Captures local defect boundaries and stress concentrations

Global branch (Multi-head self-attention)

- Full $N \times N$ attention over all 10,897 nodes
- Captures global mode shapes and structural coupling

Fusion: gated combination of both representations



Memory: $O(N^2)$ global attention requires batch size = 1 on 24 GB GPU.

Results

(A) Static FEM — architecture & PINN comparison (interim ep 40–100)

Model	Task	OptF1	AUC	deb	fod	imp	del	Reg. RMSE
LGSTA	NC+L1+L2	**0.862**	0.999	0.426	0.414	0.357	0.277	—
SAGE	NC+L2	0.845	0.999	0.298	0.381	0.404	0.002	—
LGSTA	MT+L1+L2	0.837	0.999	0.361	0.388	0.363	0.241	**0.8 mm**
GATv2	MT+L1+L2	0.829	0.999	0.342	0.315	0.359	0.000	0.7 mm
SAGE	MT+L1+L2	0.826	0.999	0.324	0.394	0.348	0.182	0.7 mm
GCN	MT+L1+L2	0.819	0.999	0.486	0.502	0.082	0.584	0.6 mm
GIN	MT+L1+L2	0.812	0.998	0.186	0.000	0.260	0.030	1.2 mm
Transolver	MT+L1+L2	0.010	0.493	—	—	—	—	—

NC = node cls only   **MT** = +centroid regression   **L1** = stress-gradient PINN   **L2** = equil. residual PINN

Interim results (ep 40–100); all runs continue to 200 epochs.

(B) Dynamic GW — MIFNO-GW v2 (sensor-consistent)

Model: MIFNO-GW (waveform-only FNO)

2-D factorised FNO — Lehmann et al. 2025

Input: 9 sensors × 4,000 time steps

Task: binary defect / healthy classification

	Train	Val
Samples	86	22
Healthy	81	20
Defect	5	2

Training: ep 200, $\lambda_{\text{disp}} = 0.05$, results pending (runs on Vancouver)

Data quality fix (v1 → v2)

Root cause found: healthy CSVs (9 sensors, circumferential ring) and defect CSVs (99 sensors, axial line) used *different* physical sensor layouts. v1 F1 = 1.0 was a position-leakage artifact — the model learned to distinguish sensor-grid patterns, not wave signatures.

Fix: filter to 9-sensor position-consistent subset (108 samples); remove position branch from model (waveform-only FNO).

Level-3 PINN — wave phase consistency

$$\mathcal{L}_{\text{disp}} = \left\langle w_{ij} \left[\angle(\hat{u}_j \hat{u}_i^*) - \frac{\omega |dx|}{c} \right]^2 \right\rangle_{ij,\omega}$$

Key findings

(A-1) LGSTA wins at OptF1 = 0.862

Dual local-global attention: local GAT finds defect boundaries; global self-attention captures full structural coupling.

(A-2) Multitask centroid regression: 0.6–1.2 mm RMSE

Shared backbone simultaneously classifies defect type *and* locates centroid — sub-mm accuracy on a 5 m structure.

(A-3) L2 PINN alone (SAGE, 0.845) competitive with L1+L2

$\nabla \cdot \sigma = 0$ residual is the dominant physics term; stress-gradient L1 adds marginal gain.

(A-4) Transolver collapses (AUC $\approx$ 0.49)

Designed for structured grids; fails on irregular FEM mesh even without physics loss.

(B) GW dataset: position-leakage fixed; still small

v1 F1 = 1.0 was artifact (sensor layout $\neq$ between healthy/defect). v2: 108 clean samples, waveform-only FNO. Need $\gtrsim$50 defect for L3 PINN

Outlook

Next steps

- **200-epoch results** — GCN, GIN, LGSTA, GATv2, MeshGNN convergence (target: 2026-06-23)
- **GW dataset unification** — ONE Healthy Abaqus job (99-sensor) + augment ⇒ 197-sample consistent dataset; enables L3 PINN & MIFNO-GW v3
- **Sim-to-real (OGW)** — Open Guided Waves omega-stringer dataset (18 GB, Zenodo 5105861): CFRP plate + omega stringer, 3 scenarios (intact / local / large debond), 50 kHz SLDV full wavefield. Run MIFNO-GW inference on virtual sensor traces extracted from 233 k scan points.
- **Cross-structure** — OOD: train fairing, test on stringer plate geometry
- **Fused static+dynamic** — concatenate LGSTA node logits with MIFNO-GW defect probability

Target for WCCM meeting (2026-06-23)

Final results table with all 8 architectures at 200 epochs + ablation: NC vs. MT, L1 vs. L2 vs. L1+L2.